

Angel Chavira

505-985-0411 | njlchvr@gmail.com | [Website](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

- University of Michigan - Ann Arbor, MI**Aug 2020 – Jan 2026
Ph.D. in Nuclear Engineering and Radiological Sciences (GPA: 3.97/4.00)
- University of Michigan - Ann Arbor, MI**Aug 2020 – May 2022
M.S. in Nuclear Engineering and Radiological Sciences (GPA: 3.96/4.00)
- New Mexico Tech - Socorro, NM**Aug 2016 – May 2020
B.S. in Mechanical Engineering (GPA: 3.86/4.00)

TECHNICAL SKILLS

Programming Languages:	FORTRAN, Python, C++
Modeling Methods:	Molecular dynamics, kinetic Monte Carlo, NEB, Saddle Point Search
HPC & Parallel Computing:	OpenMP, OpenMPI, TAU, DDT, GDB, MAP
Simulation & Analysis:	LAMMPS, SRIM/TRIM, OVITO
Development Tools:	Git, GitHub, Conda, CMake, Make, Agile/Scrum

RESEARCH EXPERIENCE

- University of Michigan Coupled Extremes Group: Ph.D. Student** May 2023 – Jan 2026
- Characterized the multiphysics of irradiation-creep mechanisms in Fe and Ni systems under simultaneous thermal, stress, and radiation loading, providing deeper insight into creep-mode failure and radiation-induced segregation mechanisms in novel tapered-specimen experiments.
 - Carried out multiscale simulations of point defect kinetics, diffusion under stress/temperature gradients, and microstructural degradation, spanning atomistic molecular dynamics and mesoscale kinetic Monte Carlo methods.
 - Performed 500+ point defect diffusion simulations to quantify anisotropic behavior and migration energies, finding that diffusion was strongly influenced by defect type, clustering, crystal structure, and grain orientation under stress.
 - Delivered findings that enhanced larger-scale creep-mode failure and radiation-induced segregation models, helping explain unique microstructural observations in the tapered specimens.
- University of Michigan Computational Radiation Effects Group: Ph.D. Student**Aug 2020 – Jan 2026
- Simulated microstructure evolution in electron-irradiated α -iron at experimentally relevant dose rates using a novel atomistic method, benchmarking results against molecular dynamics and experiment. Key finding: C15 Laves phase clusters play a secondary role in immobile dislocation loop formation during the aging and hardening of irradiated materials.
 - Developed an atomistic off-lattice long-time dynamics code linking atomistic defect mechanisms to mesoscale microstructural evolution in metals at extended timescales relevant to long-term nuclear materials reliability. Used saddle-point search swarms and defect optimization techniques to extend simulation timescales. Integrated LAMMPS for force evaluations and minimization functions.

- Designed and parallelized the codebase for HPC deployment using OpenMPI and OpenMP. Ran production simulations on Idaho National Laboratory's Sawtooth and Bitterroot clusters and the University of Michigan's Great Lakes and Lighthouse clusters.
- Automated discovery of defect migration energy barriers for various defect cluster configurations using saddle-point search algorithms and the NEB method via LAMMPS and Python scripting. These barrier energies served as inputs to the defect optimization technique.

Sandia National Laboratories Radiation Effects Theory Department: R&D Intern.....Jun 2022 – Sep 2022

- Contributed to verification and validation (V&V) of a Monte Carlo radiation transport code by profiling the performance of core cross-section evaluation routines.
- Optimized the radiation transport code using OpenMPI, enabling interaction cross-sections to be calculated for multiple elements simultaneously.
- Created a standard operating procedure (SOP) for reproducible code performance characterization, enabling rapid identification of performance bottlenecks.
- Participated in practical software engineering activities: worked in an Agile/Scrum development environment and followed team-implemented coding guidelines and documentation practices.

New Mexico Tech Energetic Systems Lab: Undergraduate Researcher Sep 2019 – Aug 2020

- Validated a novel tensile shock-wave failure mode in aluminum under high-strain-rate material conditions.
- Designed and ran molecular dynamics simulations of tensile shock-wave fronts propagating through bulk aluminum.
- Developed data post-processing tools to visualize density, velocity, and stress profiles over time.

New Mexico Tech Cryogenic Mechanical Testing Group: Team Lead.....Aug 2019 – May 2020

- Developed a mechanical testing rig and procedure for Jacobs to evaluate the performance of carbon-fiber coupons prepared with various epoxy solutions at cryogenic temperatures.
- Designed and built a coupon containment system in compliance with strict OSHA cryogen safety standards.

MIT Atomistic Simulation & Energy Research Group: Research AssistantMay 2019 – Aug 2019

- Generated the 'sound' of atoms from vibrational frequencies computed in LAMMPS to serve as educational material for the blind.
- Documented the workflow so that others could easily continue producing these sounds.
- Presented results at an MIT research symposium at the conclusion of the 10-week summer internship.

New Mexico Tech Telescope Truss Optimization Group: Multiphysics Lead.....Aug 2018 – May 2019

- Optimized the weight and resonant-vibration response of a space telescope truss design using COMSOL.
- Delivered a new truss design to NASA officials, decreasing the structure's mass by ~200 kg.

PUBLICATIONS

- [1] **A. Chavira**, Y. Osetskiy, F. Gao, '*Microstructural Evolution in Irradiated Ferritic Iron: Dose Rate Effects on Defect Accumulation*', Scripta Materialia (in review) <http://dx.doi.org/10.2139/ssrn.7095160>
- [2] S. Lim, J. Kennedy, **A. Chavira**, M. Hirsch, T. Wei, D. Ryu, '*The Profile of Extreme Tension Wave Front in Aluminum*', Journal of Materials Science (2023) <https://doi.org/10.1007/s10853-023-08244-6>

PRESENTATIONS

- [1] **A. Chavira**, C. Hirst, K.G. Field, F. Gao, '*Atomistic Simulation of Thermal and Irradiation Creep Mechanisms in BCC-Fe and FCC-Ni using Object Probability-Based Long-Time Dynamics*', The Minerals, Metals, and Materials Society (TMS), Columbus, OH, USA, 2025
- [2] **A. Chavira**, K.G. Field, F. Gao, '*Defect Evolution Under Irradiation and Stress in bcc-Fe and fcc-Ni: Insights from Defect Rate-Based Long-Time Dynamics*', Materials in Nuclear Energy Systems (MiNES), Las Vegas, NV, USA, 2025
- [3] **A. Chavira**, F. Gao, '*Addressing Defect Accumulation and Evolution in Structural Materials: Object Probability-Based Long-Time Dynamics*', Computer Simulation of Radiation Effects in Solids (COSIRES), Kingston, Canada, 2024
- [4] **A. Chavira**, F. Gao, '*Object Probability-Based Long-Time Dynamics for Structural Nuclear Material Applications*', Materials in Nuclear Energy Systems (MiNES), New Orleans, LA, USA, 2023

POSTERS

- [1] Y. Wang, **A. Chavira**, L. Shi, F. Gao, '*Computational Nuclear Materials Research Group*', Nuclear Engineering & Radiological Sciences Open House, Ann Arbor, MI, USA, 2023
- [2] **A. Chavira**, F. Gao, '*Frenkel Pair Accumulation in BCC Fe with Long-Time Dynamics*', Computer Simulation of Radiation Effects in Solids (COSIRES), Porquerolles, France, 2022
- [3] **A. Chavira**, C. Earnest, R. Morelli, A. Aliser, M. Gammill, '*Cryogenic Composite Testing*', New Mexico Tech, Socorro, NM, USA, 2020
- [4] **A. Chavira**, A. Rohskopf, K. Gordiz, A. Henry, '*Sonification of the Elements*', Massachusetts Institute of Technology Summer Research Program (MSRP), Cambridge, MA, USA, 2019

LEADERSHIP ACTIVITIES

New Mexico Tech American Society of Mechanical Engineers: TreasurerAug 2018 – May 2019

- Managed bookkeeping and approved financial logistics for all chapter events, projects, and membership costs.
- Raised thousands of dollars through the chapter's annual golf tournament by engaging with local engineering firms.

MENTORING PORTFOLIO

University of Michigan Nuclear Engineering Mentoring Program: MentorAug 2023 – May 2024

- Mentored a 1st-year undergraduate engineering student in navigating their academic career and working toward their professional goals.

PROFESSIONAL MEMBERSHIPS

Tau Beta Pi (TBP): Member2020 – Present
The Minerals, Metals, and Materials Society (TMS): Member2023 – Present
American Nuclear Society (ANS): Member2022 – Present
Society of Hispanic Professional Engineers (SHPE): Member2018 – Present
American Society of Mechanical Engineers (ASME): Member2017 – 2020